

JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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ABSTRACT

Counterfactual explanations (CE) make model decisions actionable by identifying input changes that attain a desired prediction, but existing generators often alter more features than are needed for the counterfactual effect. We introduce *COLA*, a model- and generator-agnostic post-hoc framework that refines a supplied CE by estimating an optimal-transport (OT) coupling between factual and counterfactual sets and using that coupling in a joint-probability-informed Shapley attribution, *p-SHAP*, to select a sparse set of edits. The theoretical analysis establishes that OT minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes and, under mild conditions, that the refined counterfactuals do not move farther from the factuals than the originals. Across four datasets, twelve models, and five CE generators, COLA preserves the target counterfactual effects while using only 26–45% of the original feature edits. On a tractable small-scale benchmark, COLA further attains near-optimal performance.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) provides the methodological basis for making artificial intelligence systems transparent, inspectable, and trustworthy (Arrieta et al., 2020; Das & Rad, 2020). Within this area, feature attributions (FA) methods, including Shapley-value approaches (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017), quantify the contribution of each input feature to a machine learning (ML) model’s output and thereby expose the variables that most strongly shape a decision. In healthcare, for example, such attributions can identify factors such as age and medical history that help clinicians interpret model behavior (Ter-Minassian et al., 2023; Nohara et al., 2022). Counterfactual Explanations (CE) (Wachter et al., 2017; Guidotti, 2022) offer a complementary form of explanation by specifying input changes that would lead to a different model outcome. The large number of proposed CE algorithms (Guidotti, 2022; Verma et al., 2020) reflects the diversity of objectives and operating assumptions in this literature: some methods construct one counterfactual per factual instance, others treat multiple observations jointly, global approaches identify dataset-level directions toward a desired output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024), and distributional methods seek a counterfactual distribution that preserves the shape and cost structure of the factual distribution (You et al., 2025). Model assumptions vary as well, from differentiability requirements to procedures specialized for tree-based or ensemble predictors.

Problem Description and Challenges This paper studies the interface between attribution and counterfactual reasoning, where the central objective is not merely to produce a counterfactual but to retain its effect with as few actionable modifications as possible. Building on the related literature summarized in Appendix A, we ask the following question: Although FA and CE are both central to interpretable and accountable AI, neither is sufficient on its own for this objective: FA may identify influential variables without specifying concrete interventions, whereas CE may provide concrete alternatives that contain unnecessary feature changes. The problem is therefore to combine the contrastive information contained in a generated counterfactual with an attribution mechanism that selects only the changes needed to preserve the counterfactual effect.

Three challenges define the methodological requirements of this problem. First, because CE objectives are task-dependent, no single generator can be expected to serve all instance-level, group-level, and

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$			$\mathbf{z}'$			$\mathbf{z}''$			$\mathbf{y}^*$
$\textcircled{Q}$	$\blacktriangleright$	$\textcircled{R}$	$\textcircled{Q}$	$\blacktriangleright$	$\textcircled{R}$	$\textcircled{Q}$	$\blacktriangleright$	$\textcircled{R}$	$\textcircled{R}$
200	5	No	250	8	Yes	200	5	No	Yes
150	3	No	150	3	No	150	7	Yes	Yes
100	2	No	350	9	Yes	100	2	No	Yes
150	6	No	150	6	No	350	6	Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] A platform seeks to increase user registrations using observed interaction data, including the amount of money spent ($\textcircled{Q}$), the number of clicks ($\blacktriangleright$), and the registration indicator ($\textcircled{R}$). In the factual data ($\mathbf{x}$), no users are registered. The action plans $\mathbf{z}'$ and $\mathbf{z}''$ modify user characteristics to move the model output toward the desired outcome ($\mathbf{y}^*$) of full registration. Both plans achieve a half counterfactual effect, but $\mathbf{z}''$ attains that effect with fewer modifications than $\mathbf{z}'$.

distributional settings. Second, a refinement procedure should avoid restrictive model assumptions, such as differentiability or a particular predictive architecture, if it is to apply broadly across black-box models. Third, feature importance scores can be misleading when they are computed independently of the counterfactual target: it is not effective to perform FA independently of CE to select the most important features to change. As shown later in Result II of Section 6, this decoupling can induce counterproductive feature modifications, or user/stakeholder actions, because globally influential features need not coincide with the specific pathways that realize the desired counterfactual outcome.

Main Contributions We make three contributions. First, we formulate and address sparse post-hoc refinement of CE: given factual data and generated counterfactuals, COUNTERFACTUALS WITH LIMITED ACTIONS (COLA) selects a limited set of feature edits that preserves the counterfactual effect while remaining independent of the underlying model and CE generator. Second, we introduce p -SHAP, a joint-distribution-informed Shapley attribution that uses an optimal transport (OT)–induced coupling between factual and counterfactual sets; under appropriate choices of the coupling, this construction recovers commonly used Shapley variants while providing a modular interface for attribution and edit selection. Third, we substantiate the framework theoretically and empirically: OT minimizes an upper bound on the W_1 divergence, refined counterfactuals are no farther from factuals than the originals under mild conditions, and across four datasets, twelve models, and five CE methods, COLA with p -SHAP achieves the same target effects with only 26–45% of the original feature edits; on a small-scale benchmark, it is near-optimal.

To our knowledge, this is the first systematic method for the problem illustrated in Figure 1 that *does not require specifying particular CE algorithms or model classes*.

2 PROBLEM FORMULATION

We formally formulate the problem described in Figure 1. Denote $f : \mathbb{R}^d \rightarrow \mathbb{R}$ as a black-box ML model. Denote by $\mathbf{x}$ any observed (factual) data with n rows and d columns ($\mathbf{x} \in \mathbb{R}^{n \times d}$, $n \geq 1$, and $d \geq 1$). Let $\mathbf{y}^*$ be the target model output ($\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$). The optimization is to look for a (group of) counterfactual data instance(s) $\mathbf{z}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$) subject to a maximum number of allowed feature changes, C , to achieve model output(s) as close as possible to $\mathbf{y}^*$. Let D denote a divergence function that measures the dissimilarity between two entities. The problem is below.

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) and the constraint equation (1b) formulate the typical CE optimization. Namely, $\mathbf{z}$ is expected to make $f(\mathbf{z})$ close to $\mathbf{y}^*$ yet stays close to $\mathbf{x}$. Then $\mathbf{z}$ can be used as a

counterpart reference to explain why $f(\mathbf{x})$ does not achieve $\mathbf{y}^*$. We do not limit the function D to any specific type of divergence function, allowing it to stay general. Example functions of D can be Euclidean distance, OT, maximum mean discrepancy (MMD), or differences of the model outcome in mean or median. Then, equation (1c)–equation (1e) compose the CE optimization constrained by actions. On top of the counterfactual data $\mathbf{z}$, we also optimize an indicator variable $\mathbf{c}$, such that z_{ik} is not allowed to change iff $c_{ik} = 0$. Maximum C changes are allowed as imposed by equation (1c). Inspecting equation (1d) and equation (1e), if $c_{ik} = 0$, x_{ik} equals z_{ik} and no changes happen at (i, k) . Otherwise, if $c_{ik} = 1$, remark that M is a sufficiently large constant such that z_{ik} has good freedom to change.

To solve equation (1), we resort to FA to identify the most influential features to obtain the modification indicator variable $\mathbf{c}$. The next section introduces commonly used Shapley value methods for FA, which, together with our later proposed one, are integrated into our algorithmic framework COLA, to obtain the refined counterfactual $\mathbf{z}$. The problem is computationally difficult even when $d = 1$ for linear models, see Appendix B.

3 PRELIMINARIES ON SHAPLEY VALUE

This section introduces commonly used Shapley value methods for FA, which, together with our later proposed one, are integrated into our algorithmic framework COLA. We first introduce the concept of Shapley value in game theory, followed by a discussion on various commonly used Shapley methods for FA. Readers could find in (Sundararajan & Najmi, 2020) for a comprehensive overview of different Shapley value methods.

In cooperative game theory, a coalitional game is characterized by a finite set of players (in our context, features), denoted by $\mathcal{F} = \{1, 2, \dots, d\}$, and a characteristic function v . This function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ maps each subset $S \subseteq \mathcal{F}$ to a real number $v(S)$, representing the total payoff that can be achieved by the members of S through cooperation, with the condition $v(\emptyset) = 0$. The Shapley value, as defined based on v later, is a fundamental concept in this framework, providing an equitable method to distribute the overall payoff, $v(\mathcal{F})$, of the grand coalition among the individual players.

Prior to defining the Shapley value, denote $\Delta(k, S)$ the incremental value produced by a player (feature) k to a coalition S , aligning with the concept of marginal contribution. This is defined as the additional payoff realized by the coalition due to the inclusion of the player k , i.e.,

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i for a player i in a cooperative game is defined in equation (3) below.

$$\phi_k = \frac{1}{d} \sum_{S \subseteq \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

The value ϕ_k is computed by averaging the normalized marginal contributions $\Delta(k, S)$ of feature k across all subsets S of the feature set excluding k . The normalization factor is the inverse of the binomial coefficient $\binom{d-1}{|S|}$, where d is the total number of features. By summing these values for all subsets and dividing by d , the equation provides a fair and axiomatic distribution of payoffs among players, reflecting each features’s contribution among the whole collection of features.

For an arbitrary data point $\mathbf{x}_i$, we use $\mathbf{x}_{i,S}$ to represent its part containing only the features in S , and $\mathbf{x}_{i,\mathcal{F} \setminus S}$ the other remaining part. The same rule applies for other notations as well. Below, we introduce commonly used Shapley value methods.

Baseline Shapley (B-SHAP). This approach relies on a specific baseline data point $\mathbf{r}_j$ as the reference for any $\mathbf{x}_i$. The set function is defined to be

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j,\mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

where a feature’s absence is modeled using its value in the reference baseline data point $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). This method assumes an exact alignment between $\mathbf{x}$ and $\mathbf{r}$.

Random Baseline Shapley (RB-SHAP). This approach is a variant of BShap and is implicitly used in (Lundberg & Lee, 2017) as well as the famous “SHAP” library. The equation is as follows.

$$v_{\text{RB}}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where $\mathcal{D}$ is the background distribution that is commonly selected to be the training dataset (Lundberg & Lee, 2017; Merrick & Taly, 2020).

Counterfactual Shapley (CF-SHAP). This is an extension from RB-SHAP, by taking the distribution $\mathcal{D}$ as the counterfactual distribution defined on every single factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$).

$$v_{\text{CF}}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

This method has been adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021), demonstrated with advantages for contrastive FA. This method assumes a probabilistic alignment between $\mathbf{x}$ and $\mathbf{r}$.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We generalize equation (4) to equation (6) by integrating an algorithm A_{Prob} that returns their joint probability.¹ Our p -SHAP is defined as follows,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

Crucially, unlike prior methods that rely on random baselines, p -SHAP solves the alignment problem by identifying the optimal coupling via OT. This reformulates feature attribution not merely as a prediction decomposition, but as a transport problem that minimizes the cost of explanation.

Remark that A_{Prob} is independent of the CE algorithm, but only depends on the generated counterfactual $\mathbf{r}$. By focusing solely on these fixed components, p -SHAP ensures consistency in FA without being influenced by the variability of different CE generation processes, which is a major difference to CF-SHAP. Contrary to common expectations, we demonstrate that OT can be more effective than relying on a counterfactual distribution defined by a CE generation mechanism as done by Albini et al. (2022), in Result II of Section 6 later.

Especially, one of the focus in this paper is to consider the OT problem (also the 2-Wasserstein divergence) defined below. And the transportation plan $\mathbf{p}_{\text{OT}}$ obtained by solving OT is used as the joint distribution of $\mathbf{x}$ and $\mathbf{r}$ in p -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Note that $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ represent the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$ respectively, and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ the set of joint distributions (i.e. all possible transport plans). The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is the entropic regularization with $\varepsilon \geq 0$ being the coefficient. Such regularization ($\varepsilon > 0$) helps accelerate the computation of OT.

Theoretical Aspects of p -SHAP OT minimizes the total feature modification cost (i.e. modifying $\mathbf{x}$ towards $\mathbf{r}$) under its obtained alignment between factual $\mathbf{x}$ and counterfactual $\mathbf{r}$. This directly corresponds to our objective of finding feature modifications that achieve the counterfactual outcomes at minimal cost. We can further strengthen this connection theoretically under the Lipschitz continuity assumption of the predictive model f . In Theorem 4.1 below (proof in Appendix C), we establish that the transportation plan $\mathbf{p}_{\text{OT}}$ used in p -SHAP is effective in minimizing an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$. Specifically, the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$, is

¹First, p -SHAP degrades to B-SHAP in equation (4) when A_{Prob} defines a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates an $i \leftrightarrow j$ alignment of for any $\mathbf{x}_i, \mathbf{r}_j$. Second, p -SHAP degrades to RB-SHAP in equation (5) when A_{Prob} is defined to be independent of CE but associates with an arbitrary distribution $\mathcal{D}$. Third, p -SHAP degrades to CF-SHAP in equation (6) when A_{Prob} is built upon a known distribution of CE.

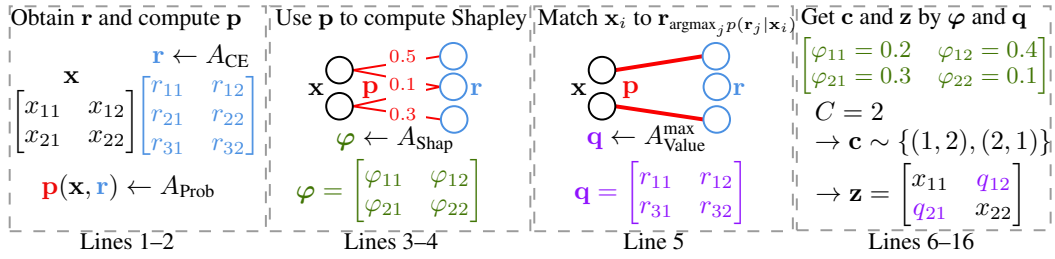


Figure 2: [An illustration of COLA] This figure shows how COLA gets $\mathbf{c}$ and $\mathbf{z}$ for equation (1). We use $A_{\text{Value}}^{\max}$ for illustration in line 5 due to its simplicity. In lines 6–16, we assume $C = 2$, and the sampling yields exactly two positions for modifications according to the probability matrix φ .

bounded by the Lipschitz constant (assuming Lipschitz continuity of f) multiplied by the square root of the minimized expected cost of changing $\mathbf{x}$ towards $\mathbf{r}$, i.e. $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ where p_{ij} ($j = 1, 2, \dots, m$) quantify how the feature values of $\mathbf{x}_i$ should be adjusted towards those of one or multiple $\mathbf{r}_j$. Practically, this means that in p -SHAP, the OT plan $\mathbf{p}_{\text{OT}}$ provides a strategy to adjust the feature values of $\mathbf{x}$ towards those of $\mathbf{r}$ in a way that minimizes the expected modification cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$. Compared to other modification plans ($\mathbf{p} \in \Pi$), $\mathbf{p}_{\text{OT}}$ yields the minimal possible cost, which in turn provides the tightest upper bound on the violation of the counterfactual effect $W_1(f(\mathbf{x}), \mathbf{y}^*)$ in proportion to this cost.

Theorem 4.1 (p -SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{\text{OT}}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{\text{OT}}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Although Theorem 4.1 bounds the transport cost, it serves as a convex proxy for the $\mathcal{NP}$ -hard discrete sparsity (L_0) problem. By concentrating attribution mass on the most efficient paths, the OT coupling effectively guides the greedy selection in Algorithm 1 to discard non-essential features. In addition, p -SHAP is conceptually correct in attributing the causal behavior to the modifications of the characteristics, stated in Theorem 4.2 below (proof in Appendix D).

Theorem 4.2 (Interventional Effect of p -SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Furthermore, we remark that p -SHAP preserves nice axioms of B-SHAP and RB-SHAP, which makes it an effective tool for attributing features. We omit the proof but refer to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) as a reference for axioms of Shapley.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch The algorithmic framework COLA, stated in Algorithm 1 below, aims to solve equation (1) and is established on four categories of algorithms. We explain Algorithm 1 in details below, along with an illustration in Figure 2.

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

Line 1 (Applying a CE algorithm to find a counterfactual $\mathbf{r}$). The CE algorithm A_{CE} takes the model f , the factual $\mathbf{x}$, the target outcome $\mathbf{y}^*$, and the tolerance ϵ as input. The algorithm returns a counterfactual $\mathbf{r}$ staying close with $\mathbf{x}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.

Line 2 (Seeking a joint distribution of $\mathbf{x}$ and $\mathbf{r}$). We use an algorithm A_{Prob} for this task, which takes $\mathbf{x}$ and $\mathbf{r}$ as input, and outputs a matrix representing the joint distribution of all n and m data points in $\mathbf{x}$ and $\mathbf{r}$, respectively. The joint distribution $\mathbf{p}$ represents an alignment relationship (or matching) between the factual rows and counterfactual rows, and we use it in Line 5 to compute the values that can be used for composing $\mathbf{z}$ later on. As discussed in Section 3, A_{Prob} can be based on OT to compute a joint distribution that yields the smallest OT distance between $\mathbf{x}$ and $\mathbf{r}$, if the alignment relationship between their rows are unknown. Otherwise, it is recommended to select a joint distribution that accurately reflects the alignment between the rows in $\mathbf{x}$ and $\mathbf{r}$.

Lines 3–4 (p -SHAP FA). We apply equation (7) to compute the shapley value for $\mathbf{x}$. The joint distribution $\mathbf{p}$ can be used here (without being forced) to properly align each row of $\mathbf{x}$ with its corresponding counterfactual rows of $\mathbf{r}$, such that the selected rows in $\mathbf{r}$ serve as the most representative contrastive reference for the row in $\mathbf{x}$. Numerically, this alignment significantly influences our contrastive FA. Then, the shapley values (as a matrix) of $\mathbf{x}$ is taken element-wisely with the absolute values and normalized (such that all values sum up to one). The resulted matrix φ forms our FA.

Line 5 (Computing feature values). The algorithm A_{Value} is used for this task, which takes the counterfactual $\mathbf{r}$ and the joint distribution $\mathbf{p}$ as input. For any row i in $\mathbf{x}$, A_{Value} selects one or multiple row(s) in $\mathbf{r}$, used as references for making changes in $\mathbf{x}$. The algorithm returns a matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$, where each element q_{ik} serves as a counterfactual candidate for x_{ik} ($\forall i, k$). Below, we introduce $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, respectively for the cases of selecting single row and selecting multiple rows.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\text{max}}$ selects the row of $\mathbf{r}$ with the highest probability.

$$\mathbf{q} = A_{\text{Value}}^{\text{max}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i), k} \text{ and } \tau(i) = \arg \max_{j=1, 2, \dots, m} p_{ij}. \quad (9)$$

The algorithm $A_{\text{Value}}^{\text{avg}}$ computes q_{ik} as a convex combination as a weighted average of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Recall that the non-negative matrix φ is normalized to have its summation being one. We could hence treat it as a policy to select the positions in $\mathbf{x}$ for value replacement (i.e. $c_{ik} = 1$), as what line 7 does. Then, for any i and k with $c_{ik} = 1$, x_{ik} gets modified to q_{ik} , and the modified matrix is then returned as $\mathbf{z}$ together with $\mathbf{c}$ forming the optimized solutions of the problem in equation (1).

By Theorem 4.1, COLA is designed to minimize the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ by modifying $\mathbf{z}$ based on feature attribution results, which identify the most important features to adjust to achieve the desired counterfactual effect. The theorem below (proof in Appendix E) demonstrates that the refined $\mathbf{z}$, produced by the COLA framework, satisfies the constraint equation (1b) in the typical scenario where $n = m$, using the Frobenius norm as the distance measure. Empirical evidence supporting the general applicability of this conclusion can be found in Table 3.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Corollary 5.2. *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Complexity of COLA Let $\mathcal{O}(M_{\text{CE}})$ be the algorithm complexity of A_{CE} . For algorithm A_{Shap} , consider using weighted linear regression to estimate Shapley values, and denote by M_{Shap} the number of sampled subsets. The complexity of COLA with respect to n, m, d , and the regularization parameter ε of entropic OT is $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used. See Appendix F for more details.

6 NUMERICAL RESULTS

This section evaluates the effectiveness of COLA in addressing the problem in equation (1), with $\mathbf{y}^* = f(\mathbf{r})$ where $\mathbf{r}$ is the counterfactual obtained from a CE method A_{CE} . We adopt four different divergence functions: OT evaluates the distance between entire distributions. MMD evaluates the divergence between the means of two distributions in a high-dimensional feature space. The absolute mean difference (MeanD) and absolute median difference (MedianD) evaluate the divergence between mean and median, respectively. The numerical results aim at showing: I) *COLA’s effectiveness for modification minimality*. II) *p-SHAP’s superior performance than other Shapley methods towards modification minimality*. III) *COLA’s near-optimal performance*.

Experiment Setup The experiments are conducted with 4 datasets for binary classification tasks, 5 CE algorithms that are designed for diverse goals, and 12 classifiers, shown in Table 1, where a combination of dataset, A_{CE} algorithm, and a model defines an “experiment scenario”.

Table 1: [Experiment Scenarios Setup] Four datasets are used to benchmark five CE algorithms over 12 models. A “scenario” is defined to be a combination of dataset, A_{CE} algorithm, and a model f .

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] The table defines 6 methods for comparisons, colored to align with Figure 3. Each method is put in an experiment scenario, as defined in Table 1, for benchmarking.

Method	The probability $\mathbf{p}$ used by A_{Value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

We briefly introduce the many A_{CE} in Table 1. DiCE and KNN are data-instance-based CE methods, which yield counterfactual(s) respectively for each factual instance. AReS and GlobeCE are group-based CE methods, which find a collection of counterfactual instances for the whole factual data as a group. The algorithm Discount treats the factual instances as an empirical distribution and seeks a counterfactual distribution that stays in proximity to it.

Table 2 defines 6 methods, where CF- p_{OT} is the proposed p -SHAP and the others are baselines. Each is put in many experiment scenarios in Table 1, benchmarked comprehensively. Each method is determined by a combination of the three algorithms A_{Prob} , A_{Value} , and A_{Shap} . For example, the first row RB- p_{Uni} uses uniform distribution as the algorithm A_{Prob} to compute $\mathbf{p}$, and then the computed $\mathbf{p}$ is sequentially used in equation (9) or equation (10) of A_{Value} for computing $\mathbf{q}$, and A_{Shap} is RB-SHAP.

These methods are carefully designed for ablation studies. First, RB- p_{Uni} differs with CF- p_{Uni} in that the latter uses CE information whereas the former does not. Second, CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} use CE in FA with different joint distributions, and we demonstrate later that the distribution computed by OT outperforms the others significantly. Third, we want to make sure that OT is useful because it informs A_{Shap} with respect to the factual-counterfactual alignment, not because of other factors, and hence a comparison of CF- p_{OT} that uses such an alignment to RB- p_{OT} that does not. Finally, CF- p_{Ect} represents a special case that each counterfactual originates from a known source, which is used as an exact factual-counterfactual alignment, making CF-SHAP also B-SHAP.

Result I: COLA achieves significant action reduction with a minor loss in counterfactual effect

In Table 3, we set a goal for $\mathbf{z}$ such that $f(\mathbf{z})$ reaches 80% or 100% counterfactual effect of $f(\mathbf{r})^2$. Observe that COLA is effective in achieving this goal, requiring significantly fewer actions in $\mathbf{z}$ (i.e., modifications of features in $\mathbf{z}$), compared to the original CE $\mathbf{r}$. Using COLA, one could expect to only resort to 13%–25% of the feature changes (calculated by $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$) to achieve the counterfactual effect of 80%. In particular, only COLA with p -SHAP (that is, CF- p_{OT}) can reach the goal of the counterfactual effect of 100%, with only 26%–45% of the feature changes in original $\mathbf{r}$. Especially, p -SHAP leads to the best actional minimality, which is analyzed in details below.

Result II: p -SHAP outperforms the other Shapley methods in achieving counterfactual effect

We provide the evaluation of different Shapley methods in equation (4)–equation (7) in Figure 3, where the x-axis is the number of allowed feature changes C and the y-axis is the term $D(f(\mathbf{z}), \mathbf{y}^*)$ in equation (1). First, RB- p_{Uni} and RB- p_{OT} perform significantly worse than the others, indicating the importance of using CE information in attributing features towards modification minimality in CE. Second, the result showing that CF- p_{OT} outperforms RB- p_{OT} demonstrates that the use of OT enhances the performance of p -SHAP specifically by providing effective factual-counterfactual alignment, rather than being influenced by other factors, due to that they only differ in A_{Shap} (see Table 2). Third, p -SHAP significantly outperforms CF- p_{Uni} and CF- p_{Rnd} . This shows the effectiveness of the joint distribution obtained in OT in using p -SHAP. Namely, merely using the counterfactual information for FA (as done by CF- p_{Uni} and CF- p_{Rnd}) is not enough, and a proper alignment (which does not necessarily mean the one defined by the exact counterfactual generation mechanism, as revealed in Figure 4 later) between factual and counterfactual must be considered.

²Note that by definition, we have $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, which represents a 100% counterfactual effect since $\mathbf{y}^* = f(\mathbf{r})$. To define a counterfactual effect 80%, consider the proportion of divergence reduced by the refined CE $\mathbf{z}$. That is, $\text{Counterfactual Effect} = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D being OT.

Table 3: [COLA for Modification Minimality] This table shows the number of modified features in $\mathbf{z}$ by each method, when $f(\mathbf{z})$ reaches 80% counterfactual effect of $f(\mathbf{r})$. The result of each method is averaged by running in 4 randomly selected scenarios in Table 1, with 100 runs in each scenario. The symbol “-” means the target counterfactual effect cannot be achieved.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	-	-	-	-
	RB- p_{OT}	5.29(± 0.09)	75.9%	-	-
	CF- p_{Uni}	-	-	-	-
	CF- p_{Rnd}	-	-	-	-
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	-	-
	RB- p_{OT}	8.55(± 0.08)	50.2%	-	-
	CF- p_{Uni}	7.01(± 0.07)	41.1%	-	-
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	-	-
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	-	-
	RB- p_{OT}	-	-	-	-
	CF- p_{Uni}	2.80(± 0.04)	34.4%	-	-
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	-	-
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	-	-	-	-
	RB- p_{OT}	-	-	-	-
	CF- p_{Uni}	-	-	-	-
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	-	-
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

We observed that COLA with p -SHAP possess good performance for many of the experiment scenarios defined in Table 1, as shown in Figure 3 as well as Table 3. A massive amount of such results are further demonstrated in Appendix H. In conclusion, p -SHAP outperforms all the other Shapley methods for the sparsity in CE.

Result III: COLA is competitive with the mixed integer linear programming (MILP) optimum on tractable benchmarks This result demonstrates the effectiveness of p -SHAP in eliminating the influence of the CE generation process by replacing the CE algorithm-dependent knowledge³ of $\mathcal{D}$ with the OT joint distribution between the factual and counterfactual data, shown in Figure 4. We benchmark the method CF- p_{Ect} using COLA, and focus on solving equation (1) with MeanD as the divergence function D . Since CF- p_{Ect} relies on a known factual-counterfactual alignment, we benchmark the effectiveness of COLA for using this alignment. The theoretical optimality of COLA in this case can be obtained by solving an MILP, see Appendix G for how the MILP formulation is derived. Note that solving MILP is computationally heavy, hence only done for *German Credit*.

We can see that CF- p_{Ect} possess a near-optimal performance using DiCE. Remark that for DiCE and KNN, the factual-counterfactual pairs are independent to each others and hence we argue that COLA is effective for instance-based CE, even though our formulation in equation (1) is a generalization for group or distributional CE. It is interesting to note that CF- p_{OT} sometimes performs better than CF- p_{Ect} , because it utilizes a more theoretically grounded approach to identify the key features that require modification, whereas CF- p_{Ect} relies on CE algorithm-dependent knowledge, which lacks solid justification on its effectiveness for FA. We notice that there is still a gap between CF- p_{Ect} , CF- p_{OT} and the optimal result in KNN. Finding the best alignment is still an open question.

³This knowledge, i.e. exact alignment between factual and counterfactual, is available only by DiCE and KNN among all CE methods considered.

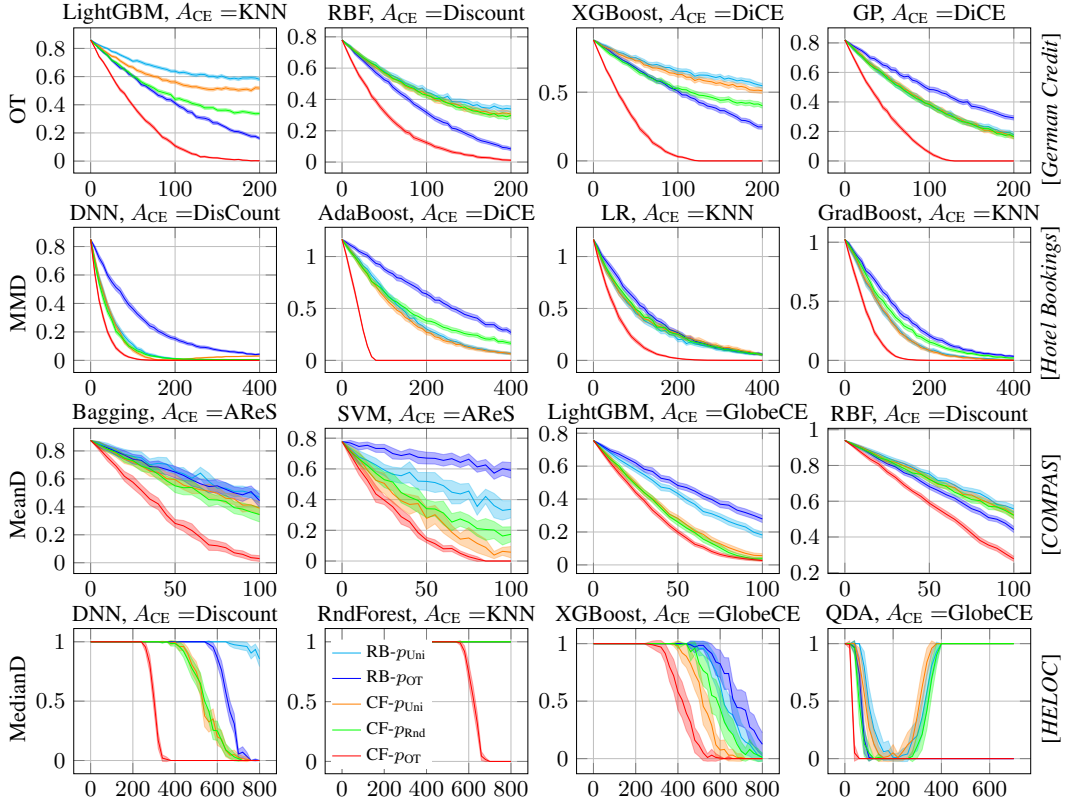


Figure 3: $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. $A_{\text{Value}}^{\text{avg}}$ is used for *HELOC* and *COMPAS*, and $A_{\text{Value}}^{\text{max}}$ is used for *German Credit* and *Hotel Bookings*. The legend inside “RndForest, $A_{\text{CE}} = \text{KNN}$ ” applies to all plots.

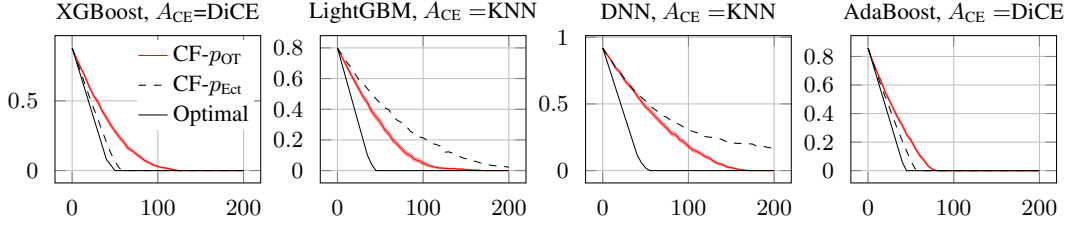


Figure 4: [*German Credit*] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C , with D being *MeanD*.

7 CONCLUSIONS

This paper introduces a novel framework, COLA, for refining CE by joint-distribution-informed Shapley values, ensuring the refined CE maintains the counterfactual effect with fewer actions.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ This paper proposes a method (COLA with *p*-SHAP) for refining CE. Our work is methodological and empirical; it does not involve human subjects or interventions, and we did not collect, annotate, or release any new personal data.

All experiments use standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC). We follow dataset licenses/terms and the common research

⁴<https://iclr.cc/public/CodeOfEthics>

practice of using these data solely for benchmarking. We do not attempt re-identification or linkage, and we release no additional personal information. Our code will include options to exclude protected attributes and to enforce feasibility/immunity constraints on features.

CE can surface biases in underlying models but do not by themselves guarantee fairness. Our method can reduce the number of recommended feature edits; it does *not* authorize edits to sensitive or immutable attributes (e.g., sex, race, age) nor does it confer causal validity. We discourage use of CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals into unrealistic or unethical behavior changes. Any deployment in consequential domains should include domain-expert-defined feasibility constraints, fairness audits (e.g., disparate impact/benefit across groups), human oversight, and compliance with applicable data-protection and AI regulations. We explicitly caution that our attribution is distributional (not causal) and must not be interpreted as causal effect.

Risks include (i) misinterpretation of attributions as causal, (ii) recommending infeasible or harmful edits, and (iii) enabling strategic manipulation. We mitigate these by (a) stating non-causal scope and limitations in the paper, (b) supporting cost/feasibility masks and immutable features in the implementation, (c) reporting performance across subgroups where relevant, and (d) providing guidance for responsible use in documentation.

We provide an anonymized implementation, detailed settings, and seeds to facilitate replication; we report model types, hyperparameters, and compute budgets, and we avoid result cherry-picking by using fixed evaluation protocols. To our knowledge, there are no conflicts of interest or sensitive sponsorships related to this submission; any funding disclosures will be provided after the double-blind review.

REPRODUCIBILITY STATEMENT

We take reproducibility seriously and provide pointers to all necessary components. The COLA framework and p-SHAP attribution are specified with pseudocode and notation in the main paper (Algorithm 1; Method and Preliminaries sections), with all theoretical assumptions and complete proofs deferred to the appendix (theorems and lemmas with line-by-line proofs).

Implementation details—hyperparameters, optimization settings, environment specifications, and random seeds—are documented in Appendix I, which also describes the datasets used and the evaluation protocol (including data splits and preprocessing). Baselines (models and CE generators) and coupling choices are enumerated in the experimental sections and ablations, with their configurations mirrored in the supplemental materials. The experiment code and software will be released upon acceptance; we also report hardware and runtime in Appendix I to contextualize compute requirements, and provide ablation/sensitivity studies and the small-scale MILP benchmark setup in the appendix to support independent verification.

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A COMPARISON TO EXISTING APPROACHES

The authors in (Albini et al., 2022) proposed CF-SHAP, which uses counterfactual data points as the background distribution for Shapley. Yet, it assumes the counterfactual data distribution is defined conditionally on each single data instance, which implies that there is a known probabilistic alignment between every factual and every counterfactual instance. Similar assumptions are made in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). In many scenarios where global explanations are expected, this assumption fails. We note that the setup in (Albini et al., 2022) for counterfactual data distribution is a special case of ours. The authors in (Kwon & Zou, 2022) proposed WeightedSHAP, adding weights to features rather than treating them as contributing equally. Our proposed method weights the contributions for data points and can be straightforwardly extended to consider weighting both rows and columns. Literature (Kommiya Mothilal et al., 2021) establishes a framework for utilizing both FA and CE for explainability. Yet, the CE-based FA have the same assumption as (Albini et al., 2022), making it difficult to generalize to group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025) cases. Another relevant work is (Sharma et al.), a method that models counterfactuals using a system’s causal structure and time-series predictors to fairly decompose a single observed change in a system-level metric into contributions from individual inputs. More importantly, the aforementioned literature does not address the minimal actions CE problem, which is the focus of our paper.

The problems formulated in (Kanamori et al., 2022; Karimi et al., 2021) are quite close to the one investigated in this paper. In (Karimi et al., 2021) the authors minimize the cost of performing actions with assumption of known structural causal model (SCM), which is rarely known in practice. The authors in (Kanamori et al., 2022) pointed out that it remains open whether existing CE methods can be used for solving that problem. An-MILP-solvers based approach is proposed for linear classifiers, tree ensembles, and deep ReLU networks, built upon the works (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). However, solving MILP is costly, which makes it difficult to scale.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 below states that one does not expect a scalable exact algorithm for solving it generally, unless $\mathcal{P} = \mathcal{NP}$, and the hardness lies not only on the non-linearity of f , but also its combinatorial nature.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. Consider the *Sparse Regression (SR) with a Cardinality Constraint* problem, defined as follows. Given a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the goal is to find a vector $\mathbf{z} \in \mathbb{R}^n$ that minimizes the residual error while having at most K non-zero

elements:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ denotes the number of non-zero elements in $\mathbf{z}$. This problem is known to be $\mathcal{NP}$ -hard due to the combinatorial nature of selecting the subset of variables to include in the model.

We will map this SR problem to our problem equation (1) with the following settings. Let the number of features be $d = 1$ and the number of instances be n (the same as the dimension of the SR problem). Let the factual data be $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$ (the zero vector), and the target model output be $\mathbf{y}^* \in \mathbb{R}^m$ (as given in the SR problem). We define the model f as a linear function $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

For the model output, we define $D(f(\mathbf{z}), \mathbf{y}^*) = \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2$, i.e. the Euclidean distance squared. For the instances, we define $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, since $\mathbf{x} = \mathbf{0}$. We set the maximum allowed feature changes C to be equal to k (the sparsity level from the SR problem). The large constant M can be any sufficiently large positive number, for example, $M \geq \max_i |z_i|$.

Given that $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) are simplified to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

This means that if $c_i = 0$, then $z_i \leq 0$ and $z_i \geq 0$, so $z_i = 0$. If $c_i = 1$, then $z_i \in [-M, M]$. The constraint equation (1c) becomes $\sum_{k=1}^n c_i \leq k$.

Our problem equation (1) thus becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this formulation, the variables c_i indicate whether the variable z_i is allowed to change ($c_i = 1$) or not ($c_i = 0$). The constraints enforce that $z_i = 0$ when $c_i = 0$, mirroring the sparsity constraint in the SR problem. The constraint $\sum_{k=1}^n c_i \leq k$ ensures that at most k features can change, matching the sparsity level. The objective function is identical to that of SR.

Since our problem formulation directly mirrors the SR problem with a cardinality constraint, which is known to be $\mathcal{NP}$ -hard, solving Problem equation (1) is at least as hard as solving the SR problem. Therefore, Problem equation (1) is $\mathcal{NP}$ -hard even when $d = 1$, the model f is linear, and the divergence functions D are standard Euclidean distances. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the set of factual data points with associated probability weights $\mu_i \geq 0$ such that $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the set of counterfactual data points with associated probability weights $\nu_j \geq 0$ such that $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below provides that feature attributions are aligned with the expected costs of feature modifications, leading to action plans that are cost-efficient in achieving counterfactual outcomes.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Proof. Since the model f is Lipschitz continuous with constant L , for any $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$, it holds that:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides of the inequality by $p_{ij}^{OT} \geq 0$, we obtain:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing both sides over all $i = 1, \dots, n$ and $j = 1, \dots, m$, we have:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let us denote $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. The inequality then becomes:

$$E_f \leq L E_d.$$

To further bound E_d , we apply the Cauchy-Schwarz inequality. Observe that the weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$ because $\mathbf{p}_{OT}$ is a probability distribution over the joint space of $\mathbf{x}$ and $\mathbf{r}$. The Cauchy-Schwarz inequality states that for any real-valued functions a_{ij} and b_{ij} ,

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy-Schwarz inequality:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the inequality for E_f , we have:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f^5 , then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Therefore, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimizes the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$, we have

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

Hence the conclusion follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ be the joint probability of $\mathbf{x}$ and $\mathbf{r}$ obtained from A_{Prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are set to $\mathbf{x}_{i,S}$, and the features in $\mathcal{F} \setminus S$ remain distributed according to their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Therefore, the expected value of $f(\mathbf{r})$ under the intervention is:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

Remark that by the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Hence the conclusion. $\square$

The value function $v^{(i)}(S)$ captures the expected value of $f(\mathbf{r})$ when we intervene by setting the features in S to $\mathbf{x}_{i,S}$, denoted as $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. This intervention is independent of any predefined joint probability distribution $p(\mathbf{x}, \mathbf{r})$. Therefore, the expression $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ represents the combined effect of the base expected value of $f(\mathbf{r})$ and the additional causal impact of the attribution $v^{(i)}(S)$.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{value}}(\mathbf{r}, \mathbf{p})$. For the elements where $c_{ik} = 1$ (modified elements), $z_{ik} = q_{ik}$. For elements where $c_{ik} = 0$, $z_{ik} = x_{ik}$. Therefore, we can write:

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms with respect to $\mathbf{q}$ and $\mathbf{z}$ are computed as follows:

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds because of equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, it follows that:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots, we get:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

Hence the conclusion. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Proof. Under the assumptions $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ becomes a permutation matrix associated with a bijection σ . Consequently, the value assignment algorithm A_{value} (whether maximizing or averaging) assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting $\mathbf{q}$ with the permuted matrix $\mathbf{r}_\sigma$ into the general bound established in Theorem 5.1 (i.e., $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$) directly yields the result. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We perform analysis of the computational complexity of COLA as follows.

First, we analyze A_{Prob} . If the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, then A_{Prob} just constructs the matrix $\mathbf{p}$ with the prior knowledge, which takes $\mathcal{O}(n \times m)$. Otherwise, we consider solving OT to obtain $\mathbf{p}$, which, by the Sinkhorn–Knopp algorithm, takes $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Then we analyze A_{Shap} . For each subset, the model is evaluated on all n data points, leading to $\mathcal{O}(n)$ evaluations per subset. Incorporating baseline values from the reference data $\mathbf{r}$ involves replacing the values of certain features with their corresponding baseline values. This operation is $\mathcal{O}(m)$ because it requires accessing the baseline values from the reference table $\mathbf{r}$ for each of the d features. If we assume that the reference values can be precomputed and accessed in constant time, then the complexity of incorporating these values can be considered as $\mathcal{O}(d)$. The number of M_{Shap} subsets results in M_{Shap} model evaluations. Combining the above steps, the complexity of A_{Shap} is $\mathcal{O}(n \times d \times M_{\text{Shap}})$.

The normalization in line 4 of COLA takes $\mathcal{O}(n \times d)$.

To compare the complexities of the two algorithms, $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, we analyze each algorithm step-by-step. For $A_{\text{Value}}^{\text{max}}$, for each row $\mathbf{x}_i$ in the data table, we need to (1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, (2) identify the row $\mathbf{r}_j$ in the reference data with the highest probability p_{ij} , which involves $\mathcal{O}(m)$ operations per row, and (3) assign $q_{ik} = r_{\tau(i),k}$ where $\tau(i) = \arg \max_j p_{ij}$, which involves $\mathcal{O}(d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{max}}$ is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, for each row $\mathbf{x}_i$ in the data table, we need to 1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, 2) compute the sum $\sum_{j'=1}^m p_{ij'}$, which involves $\mathcal{O}(m)$ operations per row, and 3) calculate the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, which involves $\mathcal{O}(m \times d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{avg}}$ is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the entire complexity is straightforwardly $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$ due to the fact $C \leq n \times d$.

Therefore, the complexity of COLA using $A_{\text{Value}}^{\text{max}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA using $A_{\text{Value}}^{\text{avg}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence the complexity of COLA with respect to n , m , d , and the regularization parameter ε of entropic OT is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

In this section, we provide a global optimality benchmark for using a known alignment between factual and counterfactual in solving equation (1) with D being MeanD, namely

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Since COLA is used, we have $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$, and $\mathbf{z}$ stays closer to $\mathbf{x}$ than $\mathbf{r}$, hence equation (1b) is dropped. The formulation of equation (1) then becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

Note that the original constraints equation (1d) and equation (1e) merge to be equation (13c), because CF- p_{Ect} is imposed to be used. That is, for any $\mathbf{x}_i$, there is an exact $\mathbf{r}_j$ serves as its reference in A_{Shap} and A_{Value} , such that x_{ik} ($k = 1, 2, \dots, d$) either stays unchanged or can be changed to r_{jk} . Therefore $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$ of which the value depends on the binary variable c_{ik} .

Due to the known alignment between any $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$, $\mathbf{q}$ is determined (also, both $A_{\text{Values}}^{\max}$ and $A_{\text{Values}}^{\text{avg}}$ return the same $\mathbf{q}$). For any data point i and any feature set $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denotes the solution $\mathbf{z}_i$ where we have all features $k \in S$ changed to q_{ik} , and the other features $h \in \mathcal{F} \setminus S$ stays x_{ih} . Hence the set of $\mathbf{z}_{iS}$ ($S \subseteq \mathcal{F}$) composes the domain of all possible values of $\mathbf{z}$. Define a corresponding scaler variable for any $\mathbf{z}_{iS}$:

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the value of the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ can be represented by a binary variable a_{iS} together with the scaler g_{iS} , namely,

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The optimization problem equation (13) is hence reformulated as a mixed integer programming below.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing η under the two constraints equation (14b) and equation (14c) is equivalent to minimizing the objective function of equation (13). The constraints in equation (14d) guarantees that each data point i is subject to one and only one feature modification plan a_{iS} for a specific S ($S \subseteq \mathcal{F}$). The constraint equation (14e) corresponds to equation (13b). Solving equation (14) yields the theoretical optimality of COLA using a known alignment between factual and counterfactual, demonstrated in Figure 4 in Section 6.

H EXTENDED NUMERICAL RESULTS

Observing Figures 5–8, CF- p_{Uni} generally performs better than RB- p_{Uni} . Second, consider RB- p_{OT} and CF- p_{OT} that also differ only in A_{Shap} , the latter consistently outperforms the former. Hence *RB-SHAP is not suitable for FA in CE*.

We analyze how different Shapley methods affect FA, corresponding to lines 3–4 in COLA. The shapley methods can be classified into two categories: First, consider RB- p_{Uni} and CF- p_{Uni} that differ

only in A_{Shap} . Observing Figures 5–8, $\text{CF-}p_{\text{Uni}}$ generally performs better than $\text{RB-}p_{\text{Uni}}$. Second, consider $\text{RB-}p_{\text{OT}}$ and $\text{CF-}p_{\text{OT}}$ that also differ only in A_{Shap} , the latter consistently outperforms the former. Hence *RB-SHAP is not suitable for FA in CE*.

Besides FA, the other equally important step of COLA is line 5, i.e. using the joint probability $p(\mathbf{x}, \mathbf{r})$ to compose the matrix $\mathbf{q}$, telling the factual x to which direction to change its features so as to move towards the target model outcome. We observe in Figures 5–8 that $\text{CF-}p_{\text{OT}}$ consistently outperforms all other methods throughout all experiments. Note that all the three methods $\text{CF-}p_{\text{Uni}}$, $\text{CF-}p_{\text{Rnd}}$, and $\text{CF-}p_{\text{OT}}$ provide solution’s for the joint probability $\mathbf{p}$ when the exact alignment between factials and counterfactuals are unknown. Yet, their performance differ significantly. Simply knowing the CE (and its marginal distribution) is insufficient.

OT proves to be exceptionally useful when the alignment information between factual and counterfactual instances is missing or inaccurate. Even when the CE algorithm explicitly matches each factual instance to a corresponding counterfactual, it is challenging to justify that the known alignment optimizes performance. This is supported by Figure 4 in Section 6.

Note that p_{OT} does not need to be the true joint distribution of $\mathbf{x}$ and $\mathbf{r}$ from a data generation perspective. Instead, it should guide COLA to treat $\mathbf{x}$ and $\mathbf{r}$ together for both FA and CE. Furthermore, the QDA column in Figure 5 shows stableness of OT-based methods, while others diverge significantly from the target. We emphasize that COLA, however, is *not limited to using OT as A_{Prob}* . As indicated by Figure 4, any known best $\mathbf{p}$ still has non-negligible gap to the global optimality. Devising a better A_{Prob} algorithm is hence an interesting topic worth exploration.

I EXPERIMENTS REPRODUCIBILITY

The experiments are conducted on a high performance computing (HPC) cluster, running with four nodes (for the four datasets) in parallel, with each node equipped with two Intel Xeon Processor 2660v3 (10 core, 2.60GHz) and 128 GB memory. The experiment runs approximately 5-10 hours in each node, dependent on the size of the dataset. It is also possible to reproduce the experiment on a laptop, while it costs more computational time generally than using an HPC cluster.

For the four datasets, the numerical features are standardized, and the categorical features follow either label-encoding or one-hot encoding. Practically, we did not observe remarkable difference between the two encoding methods in terms of COLA’s performance. The train-test split follows 7 : 3.

The optimality baseline as shown in Figure 4 is solved by Gurobi 11.0.2 ([gur](#)). In order to reproduce the optimality baseline, a license of Gurobi is required. Otherwise, we can resort to open-source operations research libraries such as Google-OR tools ([goo](#)). We remark that solving the MILP in Appendix G is computationally expensive, such that it may only apply to small scale datasets such as German Credit. If one wants to compute the optimality baseline for other datasets, then the number of used features needs to be reduced.

The hyperparameters of the models used in the experiment are specified as follows. The models Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA are scikit-learn models ([skl](#)), where all hyper-parameters are kept default. The models DNN, SVM, RBF, and LR are implemented by PyTorch ([Paszke et al., 2019](#)). The DNN has three layers. The SVM uses the linear kernel. The models XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are used by their scikit-learn interface, with all hyper-parameters kept default.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We used large language models (LLMs) only as general-purpose assist tools. Their role was limited to grammar, wording, and light copy-editing of author-written text. LLMs did not contribute research ideation, modeling choices, experimental design, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors. Any LLM-suggested phrasing was reviewed and edited before inclusion in the paper. The authors take full responsibility for all content; LLMs are not authors or contributors. This disclosure complies with the ICLR policy on LLM usage.

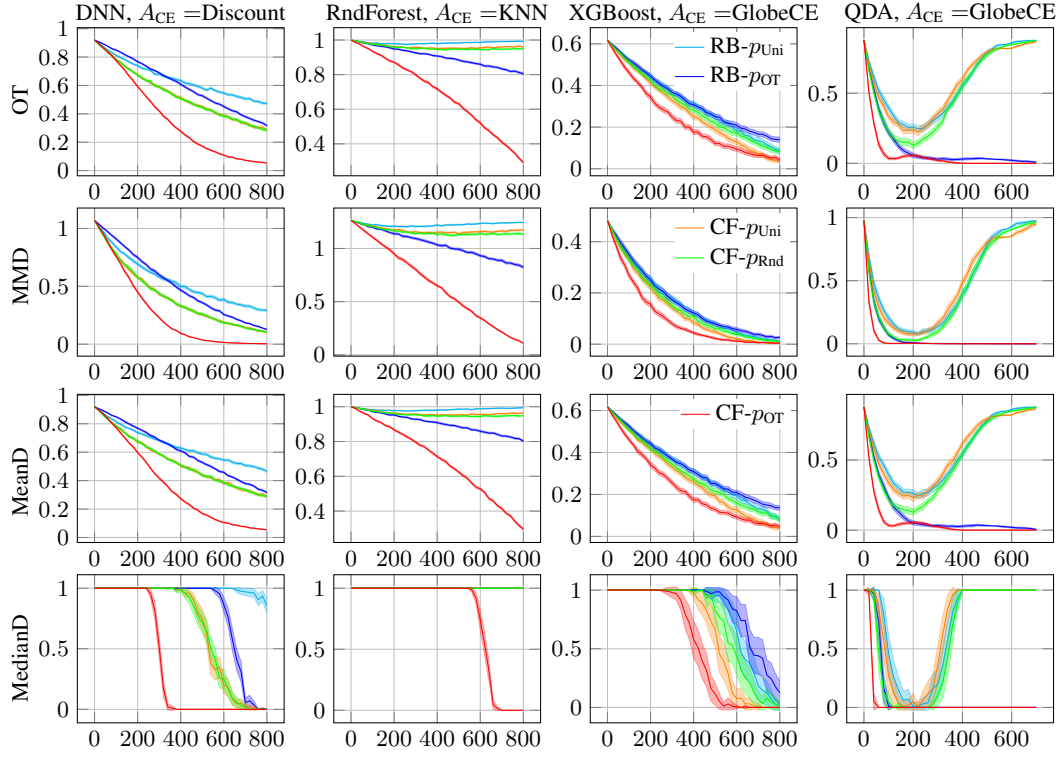


Figure 5: [HELOC] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

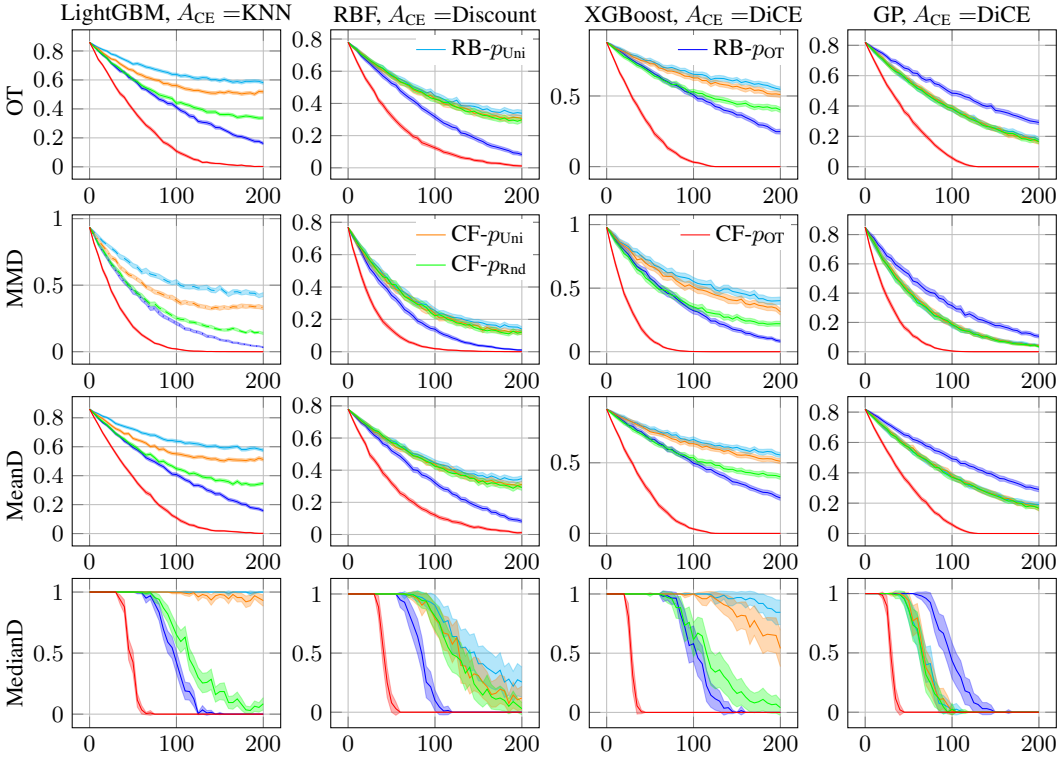


Figure 6: [German Credit] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

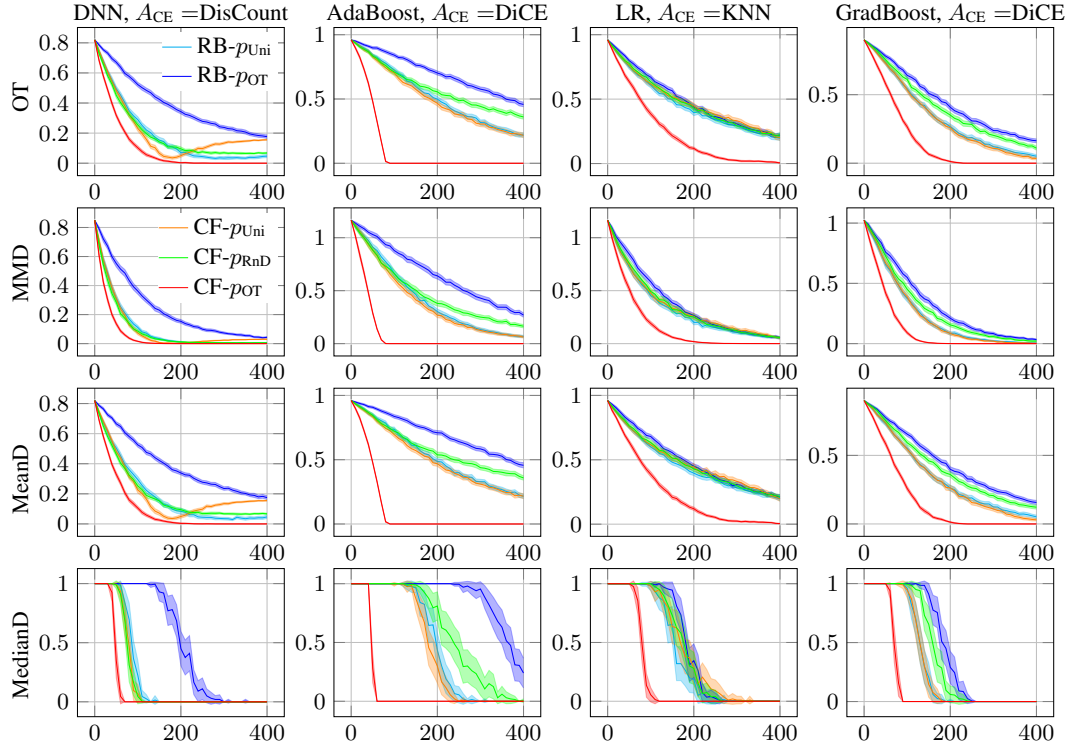


Figure 7: [Hotel Bookings] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.

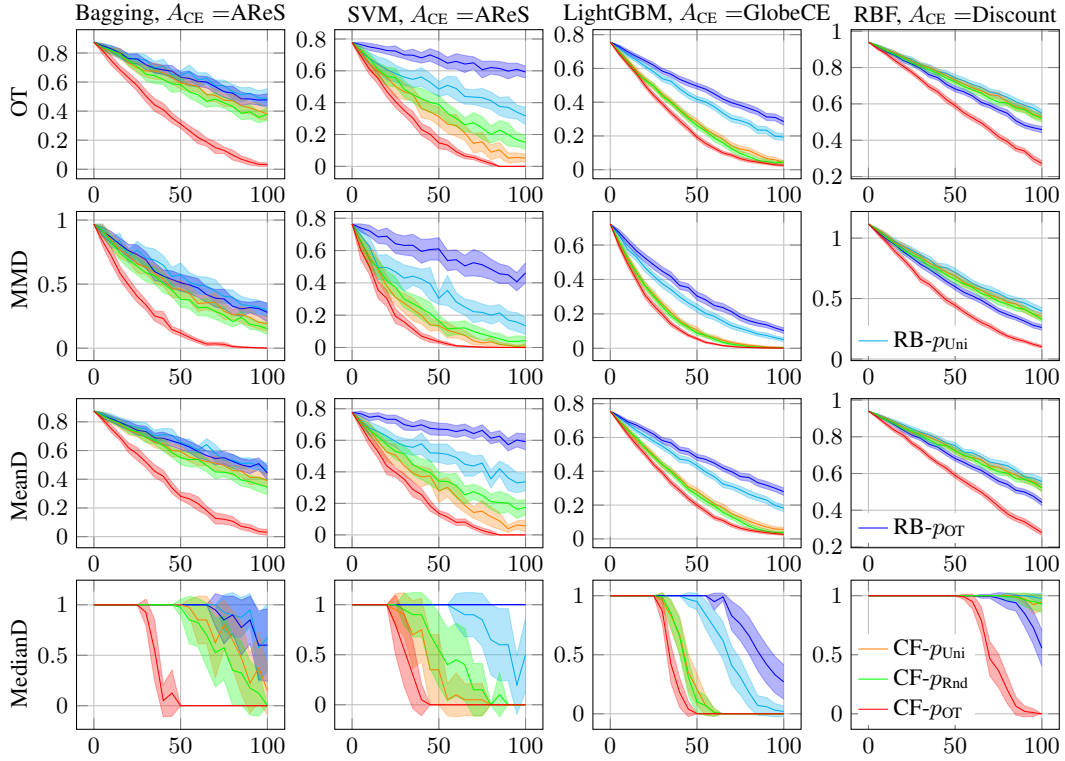


Figure 8: [COMPAS] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.